Handles Before Interventions: Access-Model UAD and the Embedded Semantics of Agency Tests

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Abstract

Unsupervised Agent Discovery (UAD) [1] is usually stated as if the observer receives variables, partitions them into internal, sensory, active, and external components, and then applies conditional-independence or control-information tests. This paper argues that this picture is too shallow for real systems. In laboratories, software audits, social systems, biological assays, recommender platforms, and human interaction, neither "intervention" nor "interface" is primitive. What the observer has are *handles*: embedded access points through which the system may be observed or operated upon. A handle has observational semantics, operational semantics, cost, latency, reversibility, spillover, and uncertain uptake. Causal graph interventions such as $\mathrm{do}(X_i = x)$ [2] are idealizations of rare depth-zero cases. We introduce an access-model formulation of UAD in which the data are generated through observation kernels and operation kernels. Agent loops, interfaces, and interventions are then inferred jointly. The resulting Bayesian/regularized objective draws on blanket-information competence scoring [3] and treats agency as posterior mass over handle-mediated control-loop hypotheses, not as a label on a raw time series. We prove an impossibility result: no method can distinguish agent structures that are equivalent under the available access model. We then give a margin-style recovery bound for finite candidate classes, an intervention-value criterion based on expected information gain over agent-loop posteriors, and a sparse layer-wise regularization scheme that encourages interpretable causal roles at every scale while allowing full agency only at selected coarse-grained bottlenecks. The framework explains why algorithmic feeds and advertising systems can manipulate humans while hiding agency: users see content handles but lack access to objective, ranking, telemetry, and policy-update handles. Interface forcing is therefore not merely convenience; it is access-channel design.

1 Motivation

UAD begins from an appealing mathematical fantasy. There is a dynamical system

$$X_t = (X_1(t), \ldots, X_n(t)),$$

and one seeks a subset $C \subseteq \{1, \ldots, n\}$ with an internal–sensory–active–external decomposition

$$C = I^C \cup S^C \cup A^C, \qquad E^C = \{1, \ldots, n\} \setminus C,$$

such that

$$J(C) := I\big(I^C_{t+1}; E^C_{t+1} \mid S^C_t, A^C_t\big) \approx 0. \tag{1}$$

The test says: conditional on sensory and active boundary variables, internal and external futures are approximately screened off.

This is a useful formal target. But it hides a prior ontological assumption: the observer has already been given the right variables at the right scale and knows enough of the interface to call some variables sensory and some active. Likewise, when we say "intervene on X_i", we often implicitly assume the Pearlean ideal [2]

$$\mathrm{do}(X_i = x),$$

as if the observer could surgically set one coordinate of the system. Real systems rarely offer such access. A chemical perturbation changes many pathways. A prompt changes a distribution over latent computations. A platform audit flag changes incentives, ranking policies, and future strategic behavior. A social request is interpreted by another agent with its own beliefs. Even a laboratory apparatus has uptake, delay, measurement error, spillover, and constraints.

Thus the primitive of intervention-capable UAD should not be a variable. It should be a *handle*. A handle is an access point with observational and/or operational semantics. The observer does not perform $\mathrm{do}(X_i = x)$; the observer applies an operation u through a handle h, producing a stochastic change in future observations:

$$u@h \rightsquigarrow P(Y_{t+1:t+H} \mid Y_{\leq t}, u, h).$$

The question becomes: given an access model, what agent loops are identifiable, and which handle-operations should be used to test them?

2 Worlds, Observations, and Handles

Let $(X_t)_{t\geq 0}$ be an unobserved world process with state space $\mathcal{X}$. The observer does not see X_t directly. Instead, observations arrive through a set of handles.

Definition 1 (Observation handle)**.** *An observation handle is a tuple*

$$h^Y = (\mathcal{Y}_h, K_h, \ell_h, \sigma_h, c_h),$$

where $\mathcal{Y}_h$ is the handle's observation space,

$$K_h : \mathcal{X} \times \mathcal{X}^- \to \Delta(\mathcal{Y}_h)$$

is an observation kernel possibly depending on current and recent world states, ℓ_h is latency, σ_h is an uncertainty or resolution descriptor, and c_h is the cost of using the handle. The observed datum is

$$Y_t^h \sim K_h(\cdot \mid X_{t-\ell_h:t}).$$

Definition 2 (Operation handle)**.** *An operation handle is a tuple*

$$h^U = (\mathcal{U}_h, \Gamma_h, \rho_h, \chi_h, c_h),$$

where $\mathcal{U}_h$ is a space of available operations,

$$\Gamma_h : \mathcal{X} \times \mathcal{U}_h \to \Delta(\mathcal{X})$$

is an operation kernel, ρ_h is reversibility, χ_h describes constraints or legality, and $c_h : \mathcal{U}_h \to \mathbb{R}_{\geq 0}$ is operation cost. Applying operation $u \in \mathcal{U}_h$ at time t induces

$$X_{t+1} \sim \Gamma_h(\cdot \mid X_t, u)$$

or, more generally, a finite-horizon kernel

$$X_{t+1:t+H} \sim \Gamma_h^H(\cdot \mid X_{\leq t}, u).$$

Definition 3 (Access model). *An access model is*

$$\mathcal{A} = (\mathcal{H}^Y, \mathcal{H}^U, K, \Gamma, \mathrm{Cost}, \mathrm{Risk}, \mathcal{C}),$$

where $\mathcal{H}^Y$ *is a family of observation handles,* $\mathcal{H}^U$ *a family of operation handles,* $K = \{K_h\}_{h\in\mathcal{H}^Y}$*,* $\Gamma = \{\Gamma_h\}_{h\in\mathcal{H}^U}$*,* Cost *and* Risk *assign costs and risks to handle-use, and* $\mathcal{C}$ *encodes constraints on allowable observations and operations.*

The data available to UAD are therefore not merely $X_{1:T}$ but

$$\mathcal{D}_T = \{(h_t^Y, y_t), (h_t^U, u_t), \mathrm{context}_t\}_{t=1}^T,$$

together with partial knowledge or priors over K and Γ.

Remark 1. *The passive time-series case is recovered by taking* $\mathcal{H}^U = \varnothing$ *and a single observation handle* h^Y *with* $Y_t = X_t + \eta_t$*. The classical UAD setting [1,4,5] is therefore a special, high-access, low-embedding case.*

3 From Variables to Handles

In classical causal notation [2], intervention is represented as graph surgery:

$$P(X_{-i} \mid \mathrm{do}(X_i = x)).$$

The handle view replaces this with an operation kernel:

$$P(Y_{t+1:t+H} \mid \mathcal{D}_{\le t}, u, h).$$

The relationship between the two is approximate and itself inferential.

Definition 4 (Effective target distribution). *For operation u through handle h, define the effective target posterior*

$$\tau_{h,u}(i) := \mathbb{P}(X_i \textit{ is directly affected by } (h,u) \mid \mathcal{D}).$$

The spillover posterior is

$$\varsigma_{h,u}(j) := \mathbb{P}(X_j \textit{ is indirectly affected by } (h,u) \mid \mathcal{D}).$$

An ideal intervention is one with concentrated τ and low ς:

$$H(\tau_{h,u}) \approx 0, \qquad \|\varsigma_{h,u}\|_1 - \max_i \tau_{h,u}(i) \approx 0.$$

Most real interventions have broad target and spillover distributions.

Definition 5 (Intervention depth). *The depth of an operation $u@h$ relative to a hypothesized target variable X_i is the minimal causal path length, in the observer's current structural posterior, from the operation event $U_t = (h,u)$ to X_i. Depth zero corresponds to* $\mathrm{do}(X_i = x)$*. Depth one corresponds to a known actuator on X_i. Depth two or greater corresponds to mediated perturbations through apparatus, environment, norms, incentives, language, or other agents.*

The depth notion is not intended as a fixed graph invariant. It is posterior and access-relative:

$$P(\mathrm{depth}(u@h, X_i) = d \mid \mathcal{D}).$$

4 Interfaces as Coupling Surfaces

A sensor/action interface is not generally a set of raw variables. It is a stable coupling surface, often constructed by an observer.

Let C be a candidate system and E its complement at some representation level. A candidate interface Q is a handle-mediated representation of coupling between C and E.

Definition 6 (Candidate interface). *A candidate interface for C is*

$$Q = (Q^S, Q^A, K_Q, \Gamma_Q),$$

where Q^S are candidate sensory handles/channels, Q^A candidate active handles/channels, K_Q describes how interface observations are produced, and Γ_Q describes how interface operations or outputs affect the world.

The interface should approximately screen off internal and external futures:

$$J_Q(C) := I(C_{t+1}; E_{t+1} \mid Q_t) \quad \text{small.} \tag{2}$$

For directed sensor/action roles, define the asymmetric information scores

$$R_S(Q; C) = I(E_t; Q_t^S \mid C_t) + I(Q_t^S; C_{t+1} \mid C_t), \tag{3}$$

$$R_A(Q; C) = I(C_t; Q_t^A \mid E_t) + I(Q_t^A; E_{t+1} \mid E_t). \tag{4}$$

These are not sufficient for agency, but they localize candidate inbound and outbound channels.

Definition 7 (Handle interface score). *For candidate C and interface Q, define*

$$\mathcal{I}(C, Q) = -J_Q(C) + \alpha_S R_S(Q; C) + \alpha_A R_A(Q; C) - \lambda_Q \,\mathrm{DL}(Q) + \mu_Q \,\mathrm{Manip}(Q). \tag{5}$$

Here $\mathrm{Manip}(Q)$ *measures whether available operation handles can selectively affect or read the interface:*

$$\mathrm{Manip}(Q) = \max_{h,u} I(u; Q_{t+1:t+H} \mid \mathcal{D}_{\le t}, h).$$

Remark 2. *Interface forcing is the deliberate construction of a high-$\mathcal{I}(C, Q)$ access channel. An API, laboratory assay, telemetry schema, randomized audit interface, or standardized social probe is not merely a measurement. It changes the access model and may change the system.*

5 Agent-Loop Hypotheses

An agent-loop hypothesis is a structured explanation of a candidate bounded process as a stateful control system.

Definition 8 (Agent-loop hypothesis). *An agent-loop hypothesis is*

$$\Omega = (C, Q, B, G, \pi, V, \Theta),$$

where C is a candidate boundary, $Q = (Q^S, Q^A)$ a candidate interface, B_t a latent belief or information state, G_t a latent goal/preference/viability state, π a policy, V a viability or objective functional, and Θ additional parameters. The model class induced by Ω predicts observations and responses by

$$B_{t+1} \sim U_\Theta(B_t, Q_{t+1}^S, Q_t^A), \tag{6}$$

$$G_{t+1} \sim R_\Theta(G_t, B_t, Q_t), \tag{7}$$

$$Q_t^A \sim \pi_\Theta(\cdot \mid B_t, G_t), \tag{8}$$

$$Y_{t+1} \sim P_\Theta(\cdot \mid B_{t+1}, G_{t+1}, Q_t^A, E_t). \tag{9}$$

The latent variables B_t and G_t need not be internal physical states. They are explanatory coordinates under which behavior compresses and interventions become predictable.

6 Bayesian Access-Model UAD

The full posterior should include both agent-loop hypotheses and access semantics:

$$P(\Omega, K, \Gamma, Q \mid \mathcal{D}_T).$$

This is the core object. It replaces both "detect an agent in a time series" and "intervene on a variable".

Definition 9 (Access-model posterior)**.** *Given priors over access semantics and loop hypotheses,*

$$P(\Omega, K, \Gamma, Q \mid \mathcal{D}_T) \propto P(\mathcal{D}_T \mid \Omega, K, \Gamma, Q)P(\Omega \mid Q)P(Q \mid K, \Gamma)P(K, \Gamma). \tag{10}$$

To make this computationally tractable, use an energy parameterization:

$$P(\Omega, Q \mid \mathcal{D}_T) \propto \exp[-E(\Omega, Q; \mathcal{D}_T)].$$

A generic energy is

$$\begin{aligned} E(\Omega, Q; \mathcal{D}_T) = \lambda_J J_Q(C) &- \lambda_C I_{\text{ctrl}}(\Omega) - \lambda_P R_{\text{persist}}(\Omega) \\ &- \lambda_B R_{\text{belief}}(\Omega) - \lambda_V R_{\text{viability}}(\Omega) - \lambda_L \Delta L_{\text{EIS}}(\Omega) \\ &- \lambda_M \operatorname{Manip}(Q) + \kappa_\Omega \operatorname{DL}(\Omega) + \kappa_Q \operatorname{DL}(Q) \\ &+ \kappa_A \operatorname{DL}(K, \Gamma). \end{aligned} \tag{11}$$

where ΔL_{EIS} is the intentional compression gain [6].

The terms have the following intended meanings:

$$I_{\text{ctrl}}(\Omega) = I(Q_t^A; E_{t+1:t+H} \mid E_t, Q_t^S), \tag{12}$$

$$R_{\text{persist}}(\Omega) = I(G_t; G_{t+k}) - \eta_G H(G_t), \tag{13}$$

$$R_{\text{belief}}(\Omega) = I(Q_t^A; B_t \mid E_t, G_t), \tag{14}$$

$$R_{\text{viability}}(\Omega) = \mathbb{E}[\Delta V_{t:t+H} \mid \pi_\Omega] - \mathbb{E}[\Delta V_{t:t+H} \mid \pi_{\text{null}}], \tag{15}$$

$$\Delta L_{\text{EIS}}(\Omega) = \log P(\mathcal{D}_T \mid M_{\text{belief/goal}}, \Omega) - \log P(\mathcal{D}_T \mid M_{\text{null}}, Q) - \lambda_{\text{int}} \operatorname{DL}(M_{\text{belief/goal}}). \tag{16}$$

Remark 3. *The non-agentic baseline should not be an unconstrained universal predictor. A powerful predictor can absorb agentic regularities without representing agent loops. Instead, the null family should consist of constrained causal topologies: passive dynamics, reactive dynamics, scripted dynamics, field dynamics, and habitual dynamics without intervention-sensitive goals.*

7 Null Models as Restricted Topologies

Let

$$\mathcal{M} = \{M_A, M_0^1, \dots, M_0^k\},$$

where M_A is an agent-loop model and M_0^i are restricted alternatives. The posterior agent probability is

$$P(M_A \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid M_A)P(M_A)}{P(\mathcal{D} \mid M_A)P(M_A) + \sum_i P(\mathcal{D} \mid M_0^i)P(M_0^i)}. \tag{17}$$

Useful nulls include:

$$M_0^{\text{passive}}: \quad X_{t+1} = F(X_t) + \eta_t, \tag{18}$$

$$M_0^{\text{reactive}}: \quad Q_t^A = f(Q_t^S) + \eta_t, \tag{19}$$

$$M_0^{\text{script}}: \quad Q_t^A = f(t, \text{context}) + \eta_t, \tag{20}$$

$$M_0^{\text{field}}: \quad X_{t+1} = F_{\text{global}}(X_t) + \eta_t, \tag{21}$$

$$M_0^{\text{habit}}: \quad Q_t^A = f(Q_{\le t}^S, Q_{<t}^A) + \eta_t, \tag{22}$$

where none includes a persistent goal variable that remains stable under changed means and predicts counterfactual policy adaptation.

8 Layered Role Regularization

It is too strong to require agentic structure at every representation layer. Low-level features may be pixels, tokens, molecules, or local dynamical modes. They need not be agentic. But they should become locally understandable.

Let a representation learner produce layers

$$Z_t^0 = X_t, \qquad Z_t^\ell = f_\theta^\ell(Z_{\leq t}^{\ell-1}).$$

At each layer ℓ, introduce soft role masks

$$m_C^\ell, m_Q^\ell, m_E^\ell, m_B^\ell, m_G^\ell \in [0,1]^{d_\ell}.$$

Use role regularization at all layers:

$$R_{\mathrm{role}}^\ell = \rho_1 R_{\mathrm{persistence}}^\ell + \rho_2 R_{\mathrm{locality}}^\ell + \rho_3 R_{\mathrm{separability}}^\ell + \rho_4 R_{\mathrm{causal-handle}}^\ell - \rho_5 R_{\mathrm{degenerate}}^\ell. \tag{23}$$

But introduce sparse agent-loop gates

$$a_C^\ell \in \{0,1\}$$

and apply the full agent score only when $a_C^\ell = 1$:

$$\mathcal{L} = \mathcal{L}_{\mathrm{pred}} - \sum_\ell \lambda_{\mathrm{role}} R_{\mathrm{role}}^\ell - \sum_{\ell,C} q(a_C^\ell = 1)\lambda_{\mathrm{agent}} R_{\mathrm{agent}}^\ell(C) + \tau \sum_{\ell,C} D_{\mathrm{KL}}\Big(q(a_C^\ell) \,\|\, p(a_C^\ell)\Big). \tag{24}$$

The prior $p(a_C^\ell = 1)$ is sparse. Most regions at most layers are not agents.

Definition 10 (Agentic structure prior)**.** *For a layer ℓ and candidate C,*

$$p(\Omega_C^\ell) \propto \exp\Big(R_{\mathrm{agent}}^\ell(C) - \kappa\,\mathrm{DL}(\Omega_C^\ell)\Big).$$

Thus agentic regularization is the log prior over latent loop structures.

9 Coarse-Graining and Hierarchical Agency

Representation layers are not necessarily ontological layers. Hierarchy should therefore be searched over coarse-graining operators

$$K_{\sigma,\tau} : \mathcal{X}_{1:T} \to \mathcal{Z}_{1:T}^{\sigma,\tau},$$

where σ denotes variable/spatial aggregation and τ temporal aggregation. Define

$$\mathrm{ScaleAgent}(C) = \mathbb{E}_{\sigma,\tau}\left[R_{\mathrm{agent}}(C; K_{\sigma,\tau})\right] - \lambda_{\mathrm{var}}\,\mathrm{Var}_{\sigma,\tau}\left[R_{\mathrm{agent}}(C; K_{\sigma,\tau})\right]. \tag{25}$$

A hierarchical agent is a stable basin in scale-space, not merely an activation in a neural-network layer.

10 Identifiability Under Access Models

The access model determines what can be known.

Definition 11 (Access equivalence)**.** *Two agent-loop hypotheses Ω and Ω' are equivalent under access model $\mathcal{A}$, written $\Omega \sim_{\mathcal{A}} \Omega'$, if for every admissible observation and operation policy ν over handles,*

$$P_\Omega(\mathcal{D}_T^\nu \mid \mathcal{A}) = P_{\Omega'}(\mathcal{D}_T^\nu \mid \mathcal{A}) \qquad \forall T.$$

The recoverable target is the equivalence class

$$[\Omega]_{\mathcal{A}} = \{\Omega' : \Omega' \sim_{\mathcal{A}} \Omega\}.$$

Theorem 1 (Access-model impossibility). *If $\Omega \sim_{\mathcal{A}} \Omega'$, then no UAD procedure whose inputs are restricted to data generated through $\mathcal{A}$ can distinguish Ω from Ω' with probability better than chance under all priors assigning equal mass to Ω and Ω'.*

Proof. By access equivalence, every admissible data-generating distribution is identical under Ω and Ω'. Hence the likelihood ratio is identically one:

$$\frac{P(\mathcal{D}_T \mid \Omega, \mathcal{A})}{P(\mathcal{D}_T \mid \Omega', \mathcal{A})} = 1.$$

With equal prior mass, posterior odds remain one for all T and all admissible operation policies. Any decision rule must therefore have symmetric error. □

Remark 4. *This theorem is the handle-level analogue of observation-channel non-identifiability. If the access model hides the objective, policy update, telemetry, or actuator channels of a platform, then user-level observations may not distinguish an optimizing recommender from a non-agentic content stream with matching distribution.*

11 Finite Candidate Recovery

Let $\mathcal{O}$ be a finite class of candidate loop/interface hypotheses. Let $S_{\mathcal{A}}(\Omega)$ be a population score, for example the negative energy in Eq. (11), and $\widehat{S}_{\mathcal{A}}(\Omega)$ its empirical estimate.

Assumption 1 (True recoverable class). *There is a true access-equivalence class $[\Omega^\star]_{\mathcal{A}}$.*

Assumption 2 (Access margin). *For all $\Omega \notin [\Omega^\star]_{\mathcal{A}}$,*

$$S_{\mathcal{A}}(\Omega^\star) - S_{\mathcal{A}}(\Omega) \geq \Delta_{\mathcal{A}} > 0.$$

Assumption 3 (Access distortion). *The implemented handles and estimated kernels distort the ideal access score by at most*

$$\sup_{\Omega \in \mathcal{O}} |S_{\widehat{\mathcal{A}}}(\Omega) - S_{\mathcal{A}}(\Omega)| \leq \delta_{\mathcal{A}}.$$

Assumption 4 (Uniform concentration). *For all $\epsilon > 0$,*

$$\mathbb{P}\left[\sup_{\Omega \in \mathcal{O}} |\widehat{S}_{\widehat{\mathcal{A}}}(\Omega) - S_{\widehat{\mathcal{A}}}(\Omega)| > \epsilon\right] \leq 2|\mathcal{O}| \exp\left(-\frac{T_{\text{eff}}\epsilon^2}{2B^2}\right).$$

Theorem 2 (Handle-recovery bound). *Let*

$$\widehat{\Omega} \in \arg\max_{\Omega \in \mathcal{O}} \widehat{S}_{\widehat{\mathcal{A}}}(\Omega).$$

If

$$\Delta_{\mathcal{A}} > 2\delta_{\mathcal{A}},$$

then

$$\mathbb{P}\left[\widehat{\Omega} \in [\Omega^\star]_{\mathcal{A}}\right] \geq 1 - 2|\mathcal{O}| \exp\left(-\frac{T_{\text{eff}}(\Delta_{\mathcal{A}} - 2\delta_{\mathcal{A}})^2}{8B^2}\right).$$

Proof. For any $\Omega \notin [\Omega^\star]_{\mathcal{A}}$,

$$S_{\widehat{\mathcal{A}}}(\Omega^\star) - S_{\widehat{\mathcal{A}}}(\Omega) \geq S_{\mathcal{A}}(\Omega^\star) - S_{\mathcal{A}}(\Omega) - 2\delta_{\mathcal{A}} \geq \Delta_{\mathcal{A}} - 2\delta_{\mathcal{A}}.$$

If empirical deviation is less than $(\Delta_{\mathcal{A}} - 2\delta_{\mathcal{A}})/2$ uniformly, no non-equivalent candidate can beat the true class. Apply the concentration assumption with this value. □

12 Intervention Selection as Handle-Operation Planning

Given posterior

$$P(\Omega, K, \Gamma, Q \mid \mathcal{D}),$$

a test is not an abstract $\mathrm{do}(X_i = x)$ but a handle-operation pair (h, u). Its value is expected posterior contraction.

Definition 12 (Expected information gain of a handle-operation)**.** *For $u \in \mathcal{U}_h$,*

$$\mathrm{EIG}(h, u) = \mathbb{E}_{y' \sim P(\cdot|\mathcal{D},h,u)} \left[D_{\mathrm{KL}}\big(P(\Omega \mid \mathcal{D}, h, u, y') \,\|\, P(\Omega \mid \mathcal{D})\big) \right]. \tag{26}$$

The recommended test is

$$(h^\star, u^\star) = \arg\max_{h,u} \left[\mathrm{EIG}(h, u) - \lambda_c \,\mathrm{Cost}(h, u) - \lambda_r \,\mathrm{Risk}(h, u) - \lambda_\chi \mathbf{1}\{(h, u) \notin \mathcal{C}\}\right]. \tag{27}$$

Proposition 1 (Diagnostic separation form)**.** *Suppose two hypotheses Ω_1, Ω_2 dominate the posterior. Let*

$$p_i^{h,u}(y') = P(y' \mid \Omega_i, \mathcal{D}, h, u).$$

Then, up to posterior weights, $\mathrm{EIG}(h, u)$ *is increasing in the predictive separation*

$$D_{\mathrm{KL}}(p_1^{h,u} \| p_2^{h,u}) + D_{\mathrm{KL}}(p_2^{h,u} \| p_1^{h,u}).$$

Thus good interventions are handle-operations for which competing loop hypotheses make different downstream predictions.

13 Agency Tests as Handle Tests

The usual intuitive agency tests become precise only when indexed by handles. Table 1 gives the translation.

Suspected structure	Handle operation	Diagnostic contrast
Sensor channel Q^S	hide, delay, corrupt, amplify input handle	behavior tracks information rather than world state
Action channel Q^A	block, attenuate, reroute output handle	environmental control collapses or compensates
Goal/resource G	move, remove, revalue resource handle	means change while latent target persists
Belief state B	false cue, occlusion, asymmetric information	action follows belief rather than actual state
Policy π	change cost, risk, payoff, time pressure	action trades off costs approximately coherently
Boundary C	separate, perturb, damage, constrain coupling	repair, compensation, or loss of autonomy
Communication	cut, corrupt, delay signal handle	coordination degrades predictably
Model of other agent	alter partner behavior or apparent belief	anticipation, teaching, deception, or adaptation
Self-maintenance	resource stress, load, disruption	actions preserve viability variables

Table 1: Agency tests rephrased as embedded handle operations.

14 Language as a High-Bandwidth Handle

Language is special because it makes latent control variables queryable. A non-linguistic observer sees trajectories:

$$Q_t^S, Q_t^A, Q_{t+1}^S, Q_{t+1}^A, \dots$$

and must infer B_t, G_t, π. A linguistic system emits public symbols that purport to describe these latents:

$$B_t \mapsto \ulcorner B_t \urcorner, \qquad G_t \mapsto \ulcorner G_t \urcorner, \qquad \pi_t \mapsto \ulcorner \pi_t \urcorner.$$

This does not guarantee truth. In adversarial settings, self-reports may be strategic. But language changes the access model by adding handles:

$$h_{\text{ask}}, \quad h_{\text{challenge}}, \quad h_{\text{explain}}, \quad h_{\text{commit}}, \quad h_{\text{counterfactual}}.$$

These handles can generate high expected information gain when the system is cooperative or only weakly adversarial.

15 Algorithmic Feeds as Hidden Agent Loops

An algorithmic feed illustrates handle failure. A user observes:

$$\text{content item} \to \text{reaction} \to \text{next content item}.$$

But the relevant loop is closer to:

$$\text{platform objective} \to \text{ranking policy} \to \text{user-state perturbation} \to \text{telemetry} \to \text{policy update}.$$

The user lacks handles for:

$$G_{\text{platform}}, \quad Q^A_{\text{ranking}}, \quad Q^S_{\text{telemetry}}, \quad \pi_{\text{update}}, \quad \text{auction and advertiser constraints}.$$

Thus user-level observations can be explained as local content variation even when a distributed optimization loop is acting on the user. Agency is hidden not because the loop is weak, but because the access model is poor.

A meaningful audit interface would expose at least:

$$\begin{aligned}&\text{candidate set}, \quad \text{ranking action}, \quad \text{objective contribution},\\ &\text{targeting features}, \quad \text{measured response}, \quad \text{policy update}.\end{aligned}$$

Such an interface changes K and Γ, increasing the recoverable margin $\Delta_{\mathcal{A}}$ and shrinking access-equivalence classes.

16 Data Contracts

Access-model UAD requires different claims for different data tiers.

Definition 13 (Tiered data contract)**.** ***Tier 1:*** *Passive access:*

$$\mathcal{D} = (Y_{1:T}, K).$$

The system can generate candidate loops but cannot make strong causal claims.

Tier 2: *Manipulable access:*

$$\mathcal{D} = (Y_{1:T}, K, \mathcal{H}^U, \Gamma, \text{Cost}, \text{Risk}).$$

The system can recommend handle-operations and update loop posteriors from their outcomes.

Tier 3: *Interface-exposed access:*

$$\mathcal{D} = (Q_t^S, Q_t^A, R_t, \mathit{update}_t, Y_{t+1})_{t=1}^T.$$

The system can audit a largely exposed control loop.

The same mathematical model can operate at all tiers, but the strength of conclusions differs:

$$\text{passive discovery} < \text{causal testing} < \text{loop auditing}.$$

17 Practical Algorithm

1. **Fit access model.** Estimate or encode observation kernels K_h, operation kernels Γ_h, costs, risks, legal constraints, and handle semantics.

2. **Learn representations.** Produce multi-scale representations $Z^{\sigma,\tau}$ through coarse-graining, prediction, and event segmentation.

3. **Propose candidates.** Use amortized heads to propose

$$q_\phi(C, Q, B, G, \pi, M \mid \mathcal{D}).$$

4. **Score candidates.** Evaluate explicit UAD/EIS/B-IQ-style energies [1, 3, 6]:

$$J_Q(C), \quad I_{\text{ctrl}}, \quad R_{\text{persist}}, \quad R_{\text{belief}}, \quad \Delta L_{\text{EIS}}, \quad \text{DL}.$$

5. **Infer posterior.** Approximate

$$P(\Omega, Q, K, \Gamma \mid \mathcal{D})$$

by variational inference, MCMC, sequential Monte Carlo, or energy-based reranking.

6. **Select test.** Choose (h, u) maximizing Eq. (27).

7. **Update.** Observe outcome y' and update posterior. Stop when posterior mass over agent-loop equivalence classes is sufficiently concentrated or when further tests are too costly.

18 Discussion

The handle approach changes the ontology of intervention-capable UAD. It says: do not start with variables and interventions. Start with access. Variables, interfaces, interventions, and intentions are all high-level abstractions over embedded dynamics.

This has several consequences.

First, agency claims become conditional:

$$\text{“agentic under access model } \mathcal{A}\text{”}.$$

No method can recover distinctions erased by $\mathcal{A}$.

Second, intervention suggestion becomes scientifically meaningful only when operation semantics are modeled. A recommendation such as “perturb variable X_i” is often ill-posed. A recommendation such as “use handle h with operation u because hypotheses Ω_1 and Ω_2 predict different outcomes” is well-formed.

Third, interface forcing is access-channel design. It may improve identifiability, but it may also distort the system. The correct question is whether the forced interface is faithful enough for the target agency claim.

Fourth, layer-wise regularization should be weakened. Lower layers should be regularized toward stable, local, separable, and manipulable roles. Full agent-loop regularization should be sparse and scale-selective.

Fifth, adversarial systems break naive EIS [6, 7]. If a system optimizes its outputs to imitate the sufficient statistics of agency while hiding its loop, behavioral compression alone is insufficient. One needs handle diversity, hidden tests, costly perturbations, or enforced audit interfaces.

19 Conclusion

Intervention, interface, and intention are all compressed names for deep embedded stacks. Causal graphs make intervention look primitive; real systems provide handles. Intentional explanations make goals look primitive; real agents expose or hide them through behavior, language, and interface structure. UAD [1] should therefore be formulated relative to an access model:

$$\mathcal{A} = (\mathcal{H}^Y, \mathcal{H}^U, K, \Gamma, \mathrm{Cost}, \mathrm{Risk}, \mathcal{C}).$$

Given $\mathcal{A}$, the target of discovery is not an agent simpliciter but a posterior over handle-mediated loop hypotheses:

$$P(C, Q, B, G, \pi, K, \Gamma \mid \mathcal{D}).$$

The recoverable object is an access-equivalence class. The appropriate test is not $\mathrm{do}(X_i = x)$ but a handle-operation chosen by expected information gain. The practical path is therefore:

$$\begin{aligned}&\text{access modeling} \to \text{role-regularized representation learning}\\&\to \text{sparse agent-loop inference} \to \text{handle-based experimental design.}\end{aligned}$$

This makes UAD less elegant but more real. It also explains why hidden optimization systems such as algorithmic feeds are hard for humans to perceive: they act through the handles we see while hiding the handles that would reveal the agent loop.

References

[1] G. Zarncke, "Foundations of unsupervised agent discovery in raw dynamical systems," *AE Studio*, 2025.

[2] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge University Press, 2009.

[3] G. Zarncke, "Bitwise intelligence: A blanket–information measure of competence," *AE Studio*, 2025.

[4] R. C. Conant and W. R. Ashby, "Every good regulator of a system must be a model of that system," *International Journal of Systems Science*, vol. 1, no. 2, pp. 89–97, 1970.

[5] M. D. Kirchhoff, T. Parr, E. Palacios, K. Friston, and J. Kiverstein, "The markov blankets of life: autonomy, active inference and the free energy principle," *Journal of the Royal Society Interface*, vol. 15, no. 138, p. 20170792, 2018.

[6] G. Zarncke, "The endogenized intentional stance: A predictive–compression formalization of agent discovery," *AE Studio*, 2025.

[7] D. C. Dennett, *The Intentional Stance*. Cambridge, MA: MIT Press, 1987.